

Students Names:
1- SAMI AWAD ALSHAMMARI
2- MOHAMMAD FUHAID ALEISSA
3- FARIS HULAYYIL ALANAZI
4- FAISAL MUDALLAH AISHARARI

Abstract

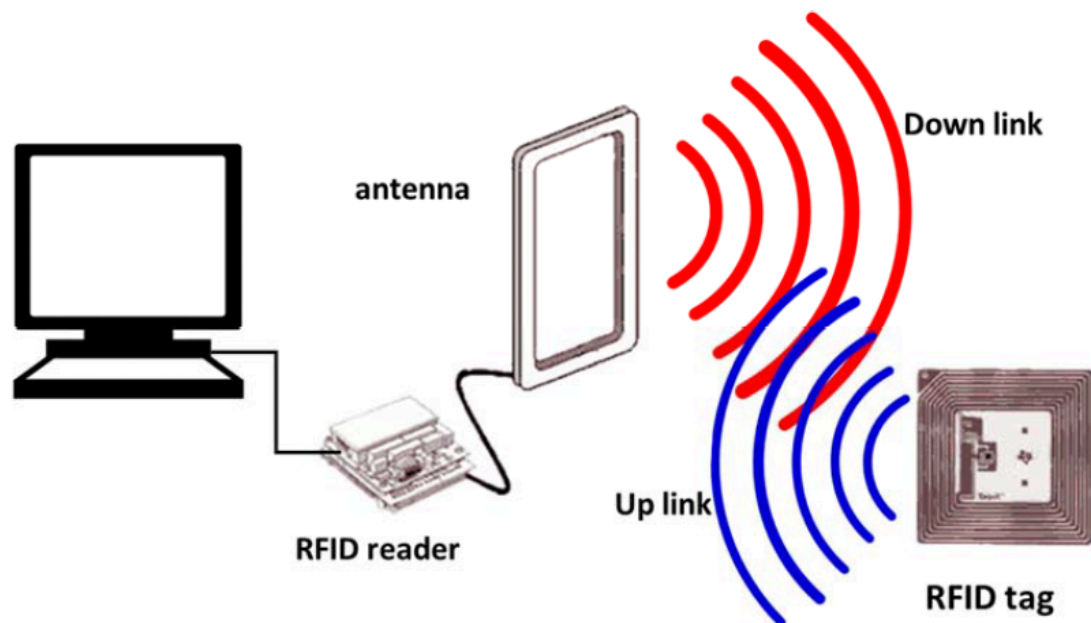
Project Title Follow Me Robot Project 2

Methodology

Supervisor Dr.Yasser Saad

Conclusion

Human following robot is very common technology and it's come a long way and has been revolutionary in assisting humans to complete tasks. The communication between the robot and the human is the most critical component to act independently. Our main goal is to design and build a robot that is not just for tracking humans, it's also to provide assistance. To make the project we used RFID Technology, a person who the robot must follow hold a RFID tag. The robot will recognize it and start follow's it.



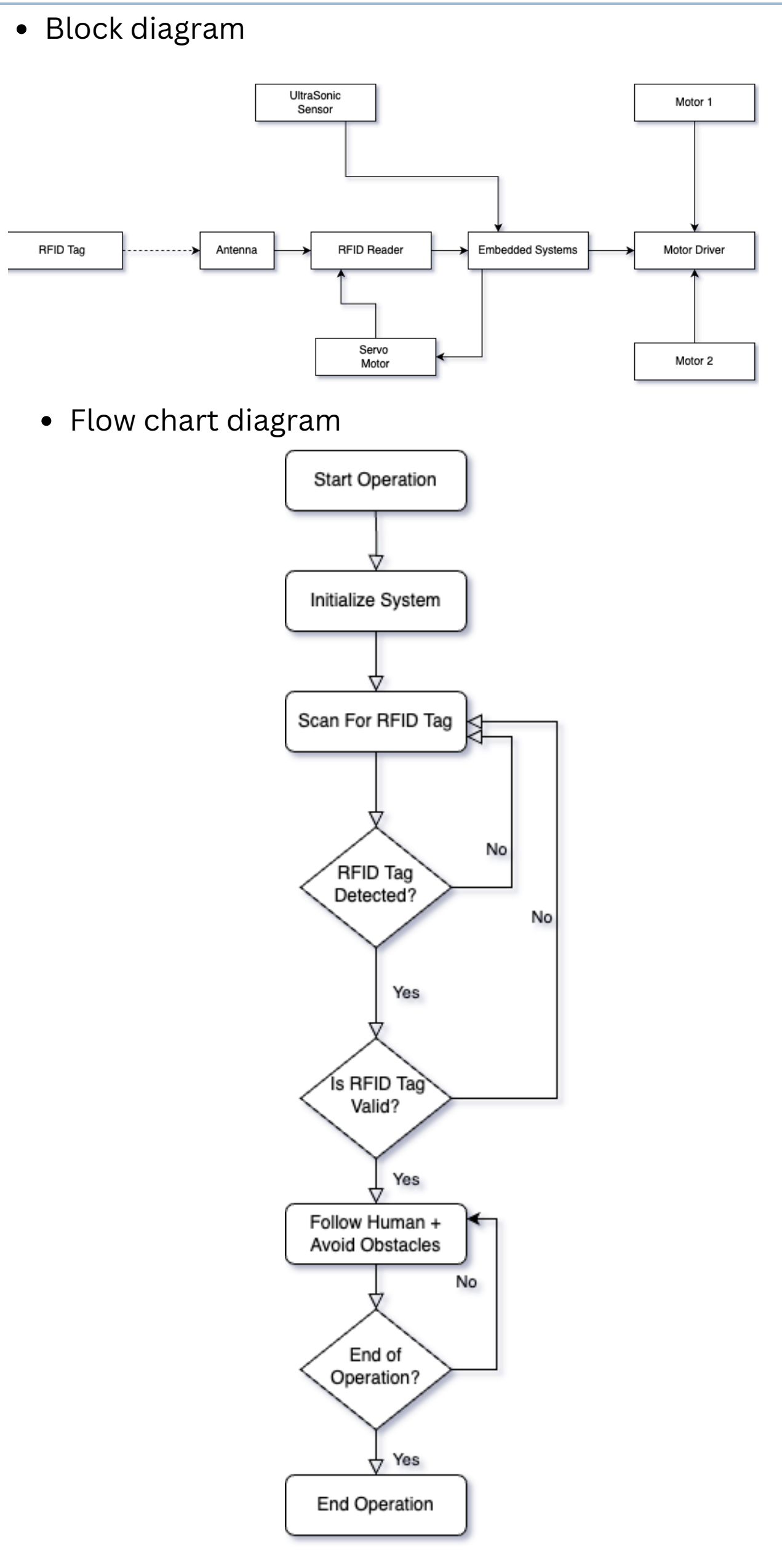
A typical RFID system consists of an interrogator, more often called a reader, transponders, also called tags, and the antennae that are connected to the reader for sending and receiving signals. Figure shows a typical RFID system. The antennae may be integrated with the reader, like a hand-held system or separate from the reader and connected by a cable. The tag is integrated with the antenna and at least one silicon chip, which responds to the reader query and stores data to built-in memory. In most cases, the reader will be connected to a computer to provide interface with other systems or human beings.

The development of a mobile robot with RFID-based human following capabilities offers an innovative solution for various applications, such as cargo handling, elderly care, and more. the combination of advanced technologies and innovative design makes this mobile robot a promising solution for addressing the challenges associated with human-robot interactions in various domains. Further research and development efforts can focus on optimizing its performance, extending battery life, and improving the robot's ability to adapt to different environments and applications.

Introduction

In the past few years, robotic technology has evolved substantially. To aid humans in their work, a robot must be able to recognize and follow them, necessitating the development of robots like the "Follow Me Robot-Assistant Robot that Follows Human", that can interact and co-exist with them. Localizing the items in the surroundings as well as the robot itself is a significant problem in enabling the robot for executing a range of activities in the real-world. A robot needs to be sophisticated to track a human through traffic regions, a vibrant atmosphere, and both indoors and out. The project's goal in building the following robot is for it to be able to assist alongside humans. It can be utilized in agriculture, construction, airports, and shopping malls. RFID-enabled human-following robots combine the unique identification capabilities of RFID technology with the versatility of autonomous robotic systems. By utilizing RFID tags to identify and track individuals, these robots can maintain a safe distance while following their designated targets, thus offering valuable assistance.

Results



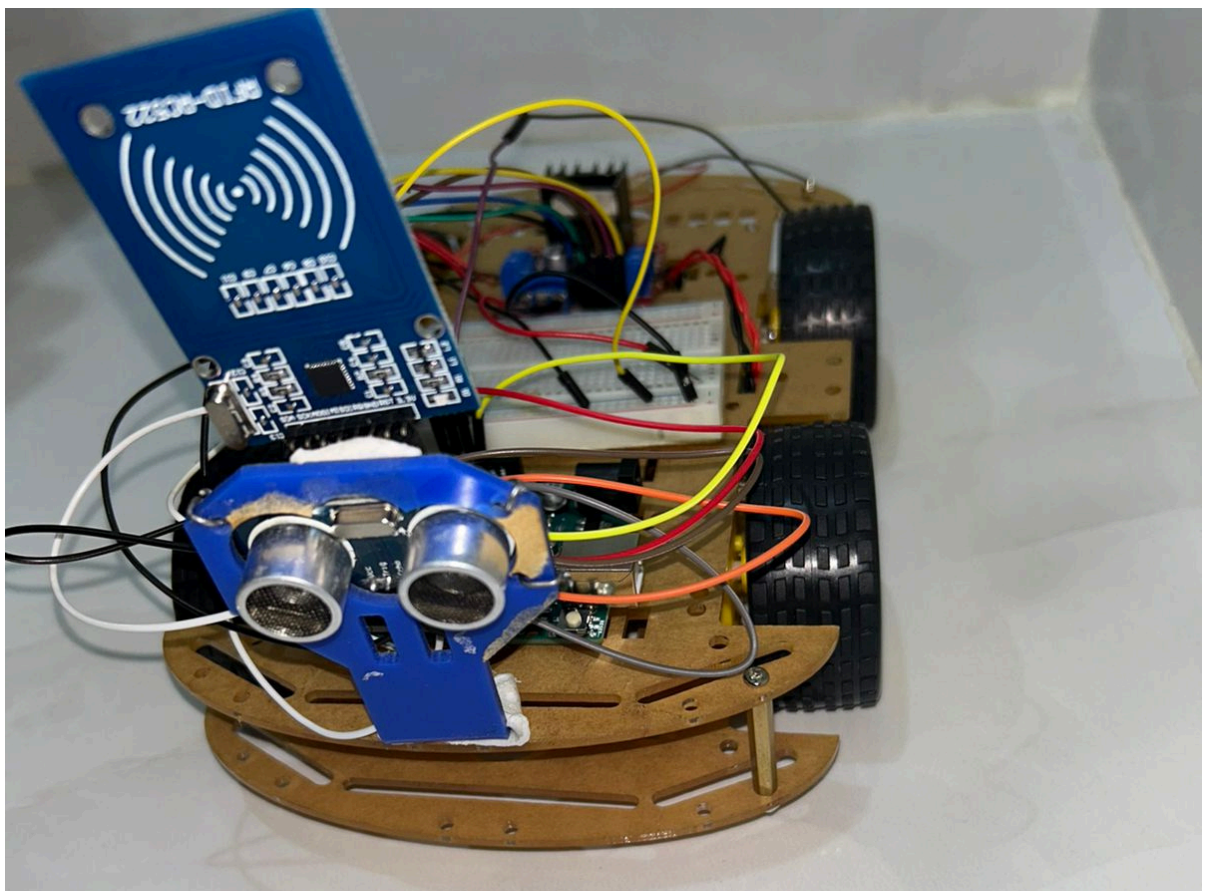
Acknowledgments

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Objectives

In the modern era characterized by rapid technological advancements, robotics has become an essential aspect of human life. The objective of project is to devise an autonomous robotic system that is capable of processing RFID tags and localizing them. The passive UHF (Ultra High Frequency) RFID technology was employed for tracking the target. The robot was designed to traverse obstacles and maintain a safe distance while following the designated individual autonomously.

Implementation



The integration of subsystems is a crucial step in ensuring the seamless functionality of the human-following robot. This phase involves combining individual hardware components and software modules to work cohesively as a unified system. In this project, the integration process focused on establishing effective communication between the RFID reader, ultrasonic sensors, motor driver, and the Arduino microcontroller to achieve real-time human tracking and obstacle avoidance.

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